Question:

Toulouse has twice as many sheep as Charleston. Charleston has 4 times as many sheep as Seattle. How many sheep do Toulouse, Charleston, and Seattle have together if Seattle has 20 sheep?

Reasoning + Answer:

<think> To solve this problem, let's first define a variable to help us work through the relationships between the cities and their sheep counts. Let's say \(S \) represents the number of sheep in Seattle. According to the problem, \(S = 20 \). So, Seattle has 20 sheep. Next, we know that Charleston has 4 times as many sheep as Seattle. So, we can express the number of sheep in Charleston (\(C \)) in terms of \(S \). Using the value of \(S \) (which is 20), we can calculate the number of sheep in Charleston as \(C = 4 \times S \). Substituting 20 for \(S \), we get \(C = 4 \times 20 \), which simplifies to \(C = 80 \). So, Charleston has 80 sheep. Now, let's focus on Toulouse. The problem states that Toulouse has twice as many sheep as Charleston. So, we can express the number of sheep in Toulouse (\(T \)) in terms of \(C \). Since \(C \) is 80, we can calculate \(T \) as \(T = 2 \times C \). Substituting 80 for \(C \), we get \(T = 2 \times 80 \), which simplifies to \(T = 160 \). Therefore, Toulouse has 160 sheep. Finally, to find out how many sheep Toulouse, Charleston, and Seattle have together, we need to add the number of sheep from all three cities: \(S + C + T \). Substituting the values we found (\(S = 20 \), \(C = 80 \), and \(T = 160 \)), we get \(20 + 80 + 160 \). Adding these together: \(20 + 80 = 100 \), and then \(100 + 160 = 260 \). </think> <answer> 260 </answer>